



Jianke Yu 俞鉴珂, PhD

PhD (2026) in Graph Machine Learning & Databases

Hangzhou, China

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RESEARCH AREAS

Machine learning for graph-structured data and database systems: graph neural network architectures for incomplete, heterogeneous and temporal graphs (fraud detection, malware detection, graph similarity computation), and learning-based methods that accelerate database and data-mining algorithms.

Keywords: Graph Neural Networks, Graph Databases, AI for Databases, Fraud Detection, Knowledge Graphs

EDUCATION

PhD, Information Technology · University of Technology Sydney 2023.07 – 2026.09

Conferred 9 September 2026. Supervised by Prof. Ying Zhang, Prof. Lu Qin and Dr Hanchen Wang. Thesis: "Fraud detection on the customer-product bipartite graph" (learning-based algorithms for graph-structured data and database systems).

M.Eng, Electronic Information · Zhejiang Gongshang University 2020.09 – 2023.06

Supervised by Prof. Xiaoyang Wang. Research on graph representation learning, fraud detection and graph similarity computation.

B.Eng, Electronic Information Engineering · Zhejiang Gongshang University 2016.09 – 2020.06

APPOINTMENTS & EXPERIENCE

Research Assistant · University of Technology Sydney 2026.07 – Present

Casual research assistant on the ARC Discovery Project led by Prof. Ying Zhang. Graph neural networks, graph similarity search and learning-based database algorithms. Ongoing.

Research Intern · Token Foundry, Alibaba Token Hub, Alibaba Group 2025.12 – Present

Research on graph learning methods for industrial data systems.

Guest Lecturer · University of Technology Sydney 2025.07 – 2025.10

Guest lecturer for UTS 43023 Emerging Topics in AI (Spring session 2025: 28 July – 24 October; UTS "Spring session" runs in the second half of the year).

Research Intern · DAMO Academy / Alibaba Cloud, Alibaba Group 2022.07 – 2023.04

Fraud detection on e-commerce platforms (work later published in KDD 2023 and IEEE TKDE).

Research Intern · Taobao, Alibaba Group 2021.11 – 2022.03

Graph-based user modelling and risk analysis.

SELECTED PUBLICATIONS (FIRST-AUTHOR CCF-A)

Overall: 19 peer-reviewed papers (10 SCI journal papers, 5 in JCR Q1; 9 conference papers) — 7 CCF-A, 6 CCF-B, 4 CCF-C; 6 as first author; 122 citations.

[1] MGDN: A Graph of Graphs Neural Network for Malware Detection. *IEEE Transactions on Knowledge and Data Engineering*, 2026. **1st author** CCF-A CAS Zone 1 Top JCR Q1

[2] SARL: A Scalable Attribute-Informed Representation Learning Framework for Bipartite Graphs. *IEEE Transactions on Knowledge and Data Engineering*, 2026. **1st author** CCF-A CAS Zone 1 Top JCR Q1

[3] Temporal Insights for Group-Based Fraud Detection on e-Commerce Platforms. *IEEE Transactions on Knowledge and Data Engineering*, 2024. **1st author** CCF-A CAS Zone 1 Top JCR Q1

[4] Group-based Fraud Detection Network on e-Commerce Platforms. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023. **1st author** CCF-A

| SCHOLARSHIPS & FUNDING

National Award for Outstanding Self-Financed Students Abroad (2025–2026) · China Scholarship Council	2026
National-level award for outstanding self-financed students abroad.	
UTS ARC Discovery Project Scholarship (Full Scholarship) · University of Technology Sydney / Australian Research Council	2023
UTS International Research Scholarship · University of Technology Sydney	2023

| HONORS & AWARDS

UTS FEIT HDR Excellence Award — Individual Category, Distinction · University of Technology Sydney, Faculty of Engineering and IT	2025
Faculty-level award (Distinction) for research excellence during HDR candidature.	
Excellent Practical Achievement Award for Professional Master Students · Education Department of Zhejiang Province	2023
Provincial award (2022 round) for outstanding professional master's practice achievements.	
Second Prize, 4th Zhejiang Province University Student Robotics Competition · Zhejiang Province	2019
Second Prize, "Internet+" Innovation and Entrepreneurship Competition · Zhejiang Gongshang University	2019

| ACADEMIC ACTIVITIES & SERVICE

Scalable Big Data Computation · Seminar, School of Data Science, The Chinese University of Hong Kong, Shenzhen 18 Apr 2026

Peer Reviewer · IEEE Transactions on Knowledge and Data Engineering (TKDE) 2025

Platform Administrator · Research Computing Platform, Prof. Ying Zhang & Prof. Xiaoyang Wang's Team, Zhejiang Gongshang University 2021–
Taking charge of the group's GPU research computing platform since its first server was set up in 2021 (training and inference environments, accounts, LDAP-based access, resource scheduling, monitoring).

| TEACHING

Guest Lecturer · 43023 Emerging Topics in Artificial Intelligence, University of Technology Sydney Spring 2025 (Jul–Oct)

Teaching Assistant · COMP9311 Database Systems, UNSW Sydney Term 3 2023

Teaching Assistant · DATA1001 Introduction to Data Science, UNSW Sydney
Term 2 2024, Term 2 2025